SENG 31242 - System Design Project (24/25)

Project - CivilCore.

Group - Data Dynamos

- SE/2022/001
- SE/2022/006
- SE/2022/018
- SE/2022/036
- SE/2022/045

Software Requirements Specification Draft

1. Introduction

1.1 Purpose and Scope

Purpose: The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for the CivilCore system. This system is designed as a simplified construction site management application for civil engineering teams, serving as a lightweight, user-friendly alternative to complex software like Primavera P6 . The primary goal is to provide supervisors, managers, and owners with a focused tool to track daily site work, materials, and overall project progress efficiently.

Scope: The CivilTrack system consists of a mobile application (built with Flutter) for on-site Supervisors, and a web application for Managers and Owners. The core capabilities of the system include:

- **Daily Reporting:** Enabling Supervisors to quickly log completed tasks, materials, labour hours, weather, and geo-tagged photos in under 5 minutes.
- **Project Setup & BOQ Integration:** Allowing Managers to set up projects, define simple Gantt-style schedules, and map Bill of Quantities (BOQ) line items directly to trackable tasks.
- **Asset & Issue Management:** Facilitating material/equipment requests from Supervisors and logging site snags with photos for Manager resolution .

- **Progress Dashboards:** Providing Managers with automated overall completion tracking against the BOQ, and giving Owners a read-only, high-level summary across all active projects.
- **Offline Functionality:** Allowing Supervisors to complete reports without an internet connection, with automatic background synchronization when connectivity is restored.

Out of Scope: The system explicitly will NOT include complex resource leveling (Primavera P6-style), financial accounting, invoicing, procurement, or vendor management functionalities.

1.2 Definitions, Acronyms, Abbreviations

- **BOQ (Bill of Quantities):** A document containing the list of work items and their planned quantities, which the system converts into trackable tasks.
- **Snag / Issue:** A physical problem or defect observed on the construction site that is flagged by a Supervisor using a geo-tagged photo for a Manager to resolve and close.
- **Gantt Schedule:** A visual project timeline consisting of planned start and end dates for each activity.
- **SRS:** Software Requirements Specification.
- **Owner:** A stakeholder who views cross-project progress and costs in a strictly read-only capacity.

1.3 Overview of Document

The remainder of this SRS document is structured as follows:

- **Section 2: Overall Description** provides a high-level overview of the system context, user personas, operating environment, and system constraints.
- **Section 3: System Analysis** contains the fact-finding summaries, system boundary diagrams, use case descriptions, and activity diagrams detailing the system workflows.
- **Section 4: Functional Requirements** enumerates the specific, testable functionalities the system must provide, traced back to our use cases.
- **Section 5: Non-Functional Requirements** outlines the performance, security, reliability, scalability, and usability constraints of the system.
- **Section 6: Alternative Solutions & Feasibility Study** evaluates different technical approaches and assesses the technical, economic, and operational feasibility of our recommended solution.

2. Overall Description

2.1 Product Perspective

The CivilCore system is an independent, cloud-based software solution designed to replace complex legacy tools like Primavera P6 for daily construction site management. The system operates across two primary interfaces: a mobile application for on-site field use and a web-based dashboard for office administration. Both platforms communicate with a central REST API backend and a MySQL database. The system is entirely self-contained for daily operational tracking but relies on cloud storage for managing project documents and geo-tagged site photos.

2.2 User Classes and Characteristics

The system caters to three distinct user personas:

- **Supervisor (Mobile App User):** Construction field workers who operate on-site. They require a highly intuitive, simplified interface as they operate in harsh environments, often with protective gear and under time constraints. Their primary technical need is a fast daily reporting workflow (under 5 minutes) and the ability to operate offline when deep inside construction zones.
- **Manager / QS (Web App User):** Office-based project managers or Quantity Surveyors who use desktop browsers. They possess moderate technical proficiency and require detailed dashboards to set up projects, import Bill of Quantities (BOQ) data, manage schedules, and monitor daily progress and budgets without making phone calls.
- **Owner (Web App User):** Executive stakeholders who require high-level visibility. Their technical interaction is minimal; they need a strictly read-only, cross-project dashboard to monitor overall progress, cost versus budget, and critical delay alerts.

2.3 Operating Environment

- **Mobile Application:** Developed in Flutter, required to run on modern iOS and Android smartphone devices.
- **Web Application:** Developed using React, accessible via standard desktop web browsers (Chrome, Edge, Firefox, Safari).
- **Backend Subsystem:** A REST API (Node.js) hosted on a cloud environment.
- **Database & Storage:** MySQL for relational data mapping (Projects, Tasks, Reports) and Cloud Object Storage for media and document libraries.
- **Network Environment:** The web application assumes a stable broadband connection. However, the mobile application explicitly assumes an unstable or non-existent cellular data connection on-site, requiring local caching and background auto-sync capabilities.

2.4 Design and Implementation Constraints

- **Time constraint:** The UI/UX must be optimized so that a Supervisor can submit a complete daily report in under 5 minutes.
- **Technology stack:** The system must utilise Flutter for the mobile application and MySQL for the database.
- **Scope limitations:** The system design must strictly avoid incorporating complex resource levelling, financial accounting, invoicing, or vendor procurement modules, keeping the focus entirely on daily site tracking.
- **Hardware limitations:** The mobile application must efficiently compress geo-tagged photos to accommodate bandwidth constraints during background synchronization.

2.5 Assumptions and Dependencies

- **Device Access:** It is assumed that all field Supervisors have access to a smartphone equipped with a functional camera and GPS module for geo-tagging.
- **Local Storage:** It is assumed that Supervisors' mobile devices have sufficient available local storage to securely cache daily reports, asset requests, and photos during offline mode.
- **Data Availability:** Subcontractor details and agreed-upon quotations are assumed to be pre-existing and entered into the system prior to supervisors logging subcontractor work.

3. System Analysis

3.1 Fact-Gathering Techniques Used

During the requirements elicitation phase, the following formal fact-finding techniques were employed to gather system requirements:

- **Structured Interviews:** Conducted with key stakeholders to understand daily site operations and project management workflows. This included interviews with a Field Supervisor and a Manager.
- **Document Analysis:** Reviewed sample Bill of Quantities (BOQ) documents (including standard Excel estimation templates and finalised physical printouts) to determine the exact data structure needed to map BOQ line items into trackable system tasks.

3.2 Existing System Analysis

The client's team currently relies on Primavera P6 for project management, alongside physical spreadsheets and paper-based printouts. **Key Pain Points Identified:**

- **Overwhelming Complexity:** Primavera P6 is too complex and feature-heavy for daily field use, presenting a steep learning curve for on-site staff.
- **Lack of Mobility:** Field supervisors lack a dedicated mobile-friendly tool to submit their daily progress directly from the construction site.
- **Manual Progress Tracking:** Managers currently struggle to track daily site progress and calculate overall completion against the BOQ without making multiple phone calls to the site.
- **Inefficient Issue Resolution:** Site problems (snags) and asset requests are often communicated informally, leading to delays and a lack of centralized tracking.

3.3 Use Case Diagram

(Please refer to `diagrams/use-case-diagram.svg` in `documents` repo for the visual representation).

The system boundary encompasses both the Mobile Application and the Web Application. The primary actors interacting with the system are:

- **Supervisor:** Operates within the mobile app boundary (submitting reports, requesting assets, logging subcontractor work, logging issues).
- **Manager / QS:** Operates within the web app boundary (setting up projects, manually entering BOQs, approving requests, viewing progress dashboards).
- **Owner:** Operates within the web app boundary (utilising a read-only, cross-project dashboard).

3.4 Use Case Descriptions

The detailed step-by-step descriptions, including main flows, alternative flows, and business rules for all system use cases, are documented individually in the `srs/use-cases/` directory in `documents` repo. The identified system use cases are:

- **UC-01:** Submit Daily Report
- **UC-02:** Setup Project and Manual BOQ Entry
- **UC-03:** Submit Asset Request
- **UC-04:** Review Asset Request
- **UC-05:** View Progress Dashboard
- **UC-06:** View Owner Dashboard
- **UC-07:** Log Subcontractor Work
- **UC-08:** Report Site Issue
- **UC-09:** Record Material Delivery
- **UC-10:** Sync Offline Data

3.5 Activity Diagrams

Activity diagrams illustrating the main and alternative workflows (including offline caching decision nodes and data validation loops) have been modelled using PlantUML. The exported SVGs and source files are available in the `diagrams/` directory in `documents` repo . Key modelled workflows include:

- `act-UC-01.svg`: Submit Daily Report
- `act-UC-02.svg`: Setup Project and Manual BOQ Entry
- `act-UC-03.svg`: Submit Asset Request
- `act-UC-05.svg`: View Progress Dashboard
- `act-UC-06.svg`: View Owner Dashboard

4. Functional Requirements

This section outlines the functional requirements for the CivilCore system. Each requirement is prioritized and traced back to its source use case to ensure complete coverage of the client's needs.

4.1 Field Operations (Supervisor App)

ID: FR-01 **Title:** Submit Daily Report **Description:** The system shall allow a Supervisor to submit a daily report consisting of completed tasks (linked to the BOQ), materials used, labour hours logged, weather conditions, and geo-tagged site photos. **Priority:** P1 - Critical **Source:** Client Brief; UC-01 **Verification:** System Test (TC-FR-01)

ID: FR-02 **Title:** Submit Asset Request **Description:** The system shall allow a Supervisor to submit a request for materials or equipment needed for the following working days, specifying the item, quantity, and required date. **Priority:** P2 - High **Source:** Client Brief; UC-03 **Verification:** Integration Test (TC-FR-02)

ID: FR-03 **Title:** Log Subcontractor Work **Description:** The system shall allow a Supervisor to log specific work completed by subcontractors on a given day and link it to the subcontractor's agreed quotation. **Priority:** P3 - Medium **Source:** Client Brief; UC-07 **Verification:** System Test (TC-FR-03)

ID: FR-04 **Title:** Report Site Issue (Snag) **Description:** The system shall allow a Supervisor to report a site issue or snag by capturing a geo-tagged photo and entering a description. The system shall set the initial status to "Open". **Priority:** P2 - High **Source:** Client Brief; UC-08 **Verification:** System Test (TC-FR-04)

ID: FR-05 **Title:** Record Material Delivery **Description:** The system shall allow a Supervisor to record the arrival of materials on-site, capturing the item, quantity, delivery date, and supplier, which updates the inventory independently of daily usage. **Priority:** P2 - High **Source:** Client Brief; UC-09 **Verification:** System Test (TC-FR-05)

4.2 Project Management (Web App)

ID: FR-06 **Title:** Project Setup and Manual BOQ Entry **Description:** The system shall allow a Manager to create a new project, assign Supervisors, define a Gantt-style schedule, and manually input Bill of Quantities (BOQ) line items to generate trackable tasks. **Priority:** P1 - Critical **Source:** Client Brief; UC-02 **Verification:** Integration Test (TC-FR-06)

ID: FR-07 **Title:** Review Asset Requests **Description:** The system shall allow a Manager to view pending asset and material requests submitted by Supervisors and update their status to "Approved" or "Rejected". **Priority:** P2 - High **Source:** Client Brief; UC-04 **Verification:** System Test (TC-FR-07)

ID: FR-08 **Title:** Calculate Progress Dashboard **Description:** The system shall automatically calculate and display the overall project completion percentage to the Manager by comparing the daily completed task quantities against the planned BOQ quantities. **Priority:** P1 - Critical **Source:** Client Brief; UC-05 **Verification:** Unit Test (TC-FR-08)

4.3 Executive Oversight (Web App)

ID: FR-09 **Title:** View Owner Dashboard **Description:** The system shall provide Owners with a read-only dashboard summarizing progress across all active projects, cost versus budget, and active alerts for delays or budget overruns. **Priority:** P2 - High **Source:** Client Brief; UC-06 **Verification:** Integration Test (TC-FR-09)

4.4 System & Connectivity

ID: FR-10 **Title:** Offline Data Synchronization **Description:** The mobile application shall cache daily reports, asset requests, and issue logs locally when operating without an internet connection, and automatically synchronize this data with the central database once a connection is restored. **Priority:** P1 - Critical **Source:** Client Brief; UC-10 **Verification:** Integration Test (TC-FR-10)

5. Non-Functional Requirements

This section specifies the non-functional requirements (NFRs) that define the system's quality attributes, performance constraints, and architectural characteristics.

5.1 Performance

- **NFR-PER-01:** The mobile application UI/UX must be optimized to ensure that a Supervisor can complete and submit a full daily report (including tasks, materials, labour, weather, and photos) in under 5 minutes.
- **NFR-PER-02:** The mobile application must efficiently compress geo-tagged photos before uploading to minimize bandwidth usage and speed up synchronization times from the field.
- **NFR-PER-03:** The Manager and Owner web dashboards must quickly aggregate and calculate overall project progress (comparing daily reports against BOQ quantities) without noticeable lag, ensuring managers can view today's progress immediately without making phone calls.

5.2 Security

- **NFR-SEC-01:** The system shall enforce strict Role-Based Access Control (RBAC). Specifically, the "Owner" role must be restricted to read-only permissions across all projects, preventing them from modifying tasks, schedules, or BOQ files.
- **NFR-SEC-02:** All data transmitted between the web application, mobile application, and the central REST API must be encrypted using secure protocols (e.g., HTTPS/TLS) to protect project data and credentials.
- **NFR-SEC-03:** The system must securely isolate project documentation and files within the cloud storage to prevent unauthorized direct access.

5.3 Usability

- **NFR-USA-01:** The system must serve as a lightweight, focused alternative to Primavera P6. The interface must be highly intuitive, specifically avoiding complex resource-levelling or financial accounting screens to eliminate steep learning curves for site staff.
- **NFR-USA-02:** The mobile application must feature large touch targets and high-contrast visuals, accommodating Field Supervisors who may be operating outdoors in bright sunlight or wearing protective gear.

5.4 Reliability

- **NFR-REL-01:** The mobile application must provide robust offline capabilities. It must reliably cache data locally on the device when operating in construction zones with zero cellular connectivity.

- **NFR-REL-02:** The system must automatically and reliably resume background synchronization of cached reports, photos, and asset requests the moment internet connectivity is restored, ensuring zero data loss.

5.5 Scalability

- **NFR-SCA-01:** The system's central database (MySQL) and REST API backend must be capable of handling concurrent daily report submissions from multiple supervisors across multiple active projects at the end of the working day.
- **NFR-SCA-02:** The system must utilize scalable cloud object storage to manage an infinitely growing library of high-resolution site photos, BOQ files, and architectural drawings without degrading database performance.

5.6 Maintainability

- **NFR-MNT-01:** The mobile application must be built using Flutter to maintain a single codebase for both iOS and Android devices, reducing future maintenance overhead.
- **NFR-MNT-02:** The backend architecture must expose a decoupled REST API (Node.js) allowing the Web (React) and Mobile (Flutter) clients to be updated and maintained independently.

6. Alternative Solutions & Feasibility Study

6.1 Alternative Solutions Analysed

Before proposing the custom CivilCore system, the following alternative solutions were evaluated:

Alternative 1: Continued Use and Customization of Primavera P6

- **Description:** Attempting to build custom mobile dashboards or simplify views within the client's existing Primavera P6 enterprise software.
- **Pros:** The software is already licensed, and historical project data is retained within a single enterprise ecosystem. It contains highly advanced resource leveling capabilities.
- **Cons:** The underlying system remains fundamentally too complex for field supervisors. It is not inherently mobile-friendly, lacks streamlined offline capabilities for harsh construction environments, and presents a steep learning curve that discourages daily use.

Alternative 2: Generic Task Management Software (e.g., Asana, Monday.com)

- **Description:** Migrating site operations to off-the-shelf, cloud-based project management tools.
- **Pros:** These tools have excellent, modern mobile applications, require minimal setup time, and are highly user-friendly.
- **Cons:** They are completely domain-agnostic and lack construction-specific data structures. They cannot natively add Bill of Quantities (BOQ) to calculate physical progress percentages, do not separate daily material usage from inventory delivery, and lack specialized workflows for subcontractor tracking and geo-tagged snag reporting.

6.2 Feasibility Study

6.2.1 Technical Feasibility The proposed system is technically highly feasible. The selected technology stack (Flutter for cross-platform mobile, React for the web dashboard, Node.js for the REST API, and MySQL for the database) consists of mature, industry-standard frameworks. Managing geo-tagged photos and documents can be reliably handled by integrating standard cloud object storage. The team possesses the required skills to implement the background offline-sync architecture required by the mobile application.

6.2.2 Economic Feasibility The project is economically feasible. By developing a custom, focused solution, the client can eliminate expensive, per-user enterprise licensing fees associated with software like Primavera P6 for their field staff. Operating costs will be restricted to predictable, scalable cloud hosting fees, which provide a high return on investment by significantly reducing manual data entry and manager phone calls.

6.2.3 Operational Feasibility Operational feasibility is exceptionally high. Field supervisors actively desire a simplified, mobile-first alternative that allows them to submit reports in under 5 minutes without dealing with complex Gantt charting or resource leveling. Because the system maps directly to their current physical workflows (BOQ task tracking, asset requests, and issue logging), user adoption will be rapid and require minimal training.

6.2.4 Schedule Feasibility The system scope has been strictly defined to exclude financial accounting, invoicing, procurement, and complex resource leveling. Because of these strict boundaries, designing the system within the 8-week SENG 31242 timeline and developing it within the subsequent 4-month SENG 34213 timeline is highly realistic and feasible.

6.3 Justified Recommendation

Based on the analysis above, we strongly recommend proceeding with the custom development of the **CivilCore system**.

Neither continuing with Primavera P6 nor adopting generic task managers fulfills the dual need for operational simplicity in the field and construction-specific BOQ tracking in the office. CivilCore strikes the exact required balance: providing supervisors with a fast, offline-capable mobile tool for daily reporting, while giving managers and owners automated, BOQ-driven progress dashboards without the overhead of enterprise accounting software.